MNC Overall Report — User Guide

This guide walks you through configuring and running the MNC Overall Report workflow, which consolidates weather observations, logistics events, wildlife sightings, and patrol data from EarthRanger into a single comprehensive monthly report.

Overview

The workflow delivers, for each run:

- **7 weather charts** (HTML + PNG) — daily precipitation, temperature, wind speed, wind gusts, soil temperature, relative humidity, and atmospheric pressure per station
- **Logistics tables** — balloon landing summary, airstrip operations (pivoted by arrival/departure), airstrip maintenance log, and airline complaints
- **Wildlife reports** — sighting maps, herd-size bar charts, and individual summary tables for elephant, buffalo, rhino, lion, leopard, cheetah, giraffe, and hartebeest
- **Livestock reports** — predation event map, predation summary, cattle count table, boma movement map and summary, illegal grazing map
- **Patrol reports** — foot, vehicle, and motorbike trajectory maps with effort summaries, overall patrol coverage map
- **1 Word report** (`overall_report.docx`) — all outputs assembled into a single document

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Prerequisites

Before running the workflow, ensure you have:

- Access to an **EarthRanger** instance with weather station observations, logistics events, wildlife sightings, livestock events, and patrol data recorded for the analysis period

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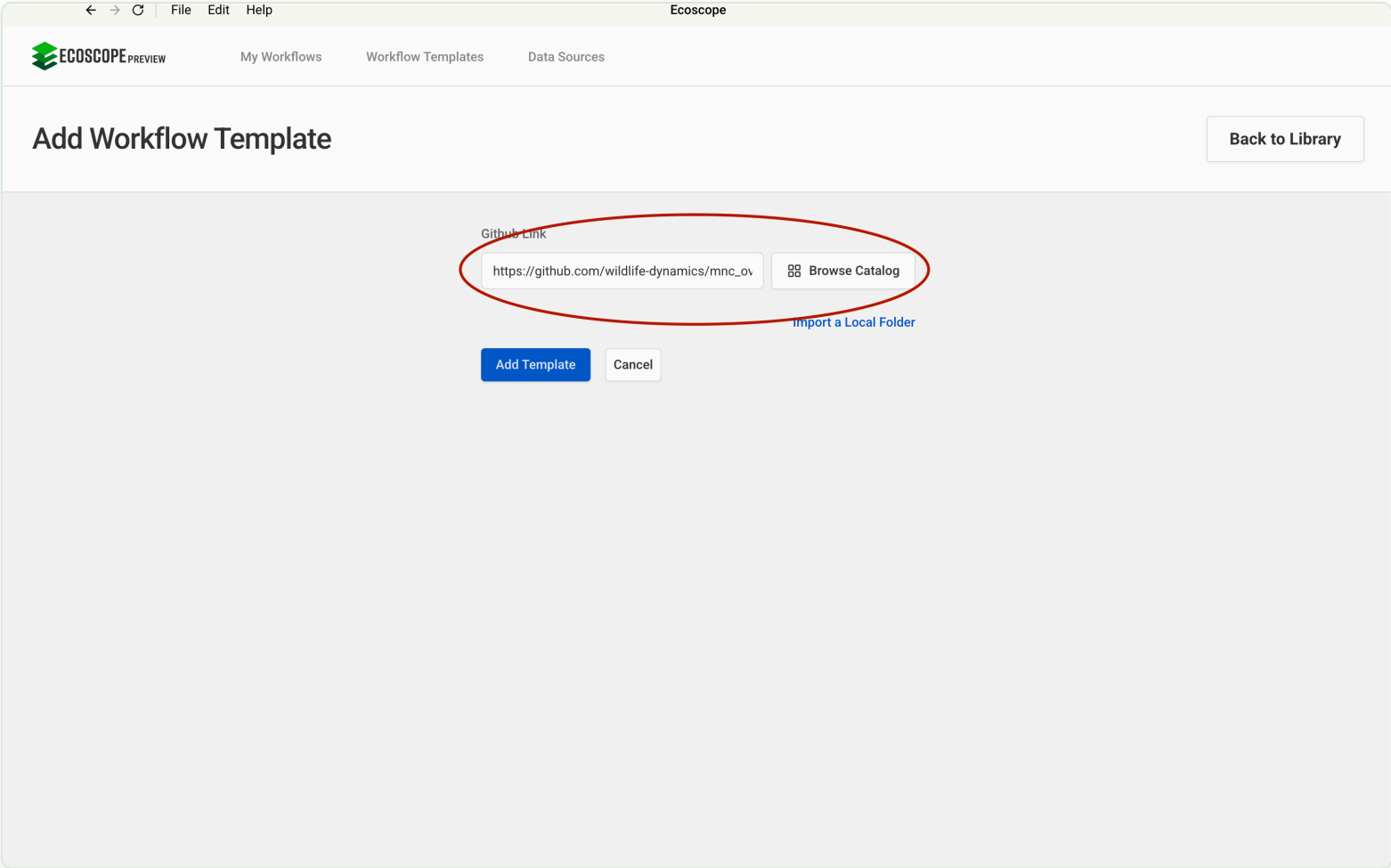
Step-by-Step Configuration

Step 1 — Add the Workflow Template

In the workflow runner, go to **Workflow Templates** and click **Add Workflow Template**. Paste the GitHub repository URL into the **Github Link** field:

```
https://github.com/wildlife-dynamics/mnc_overall_report.git
```

Then click **Add Template**.



Add Workflow Template

Step 2 — Configure the EarthRanger Connection

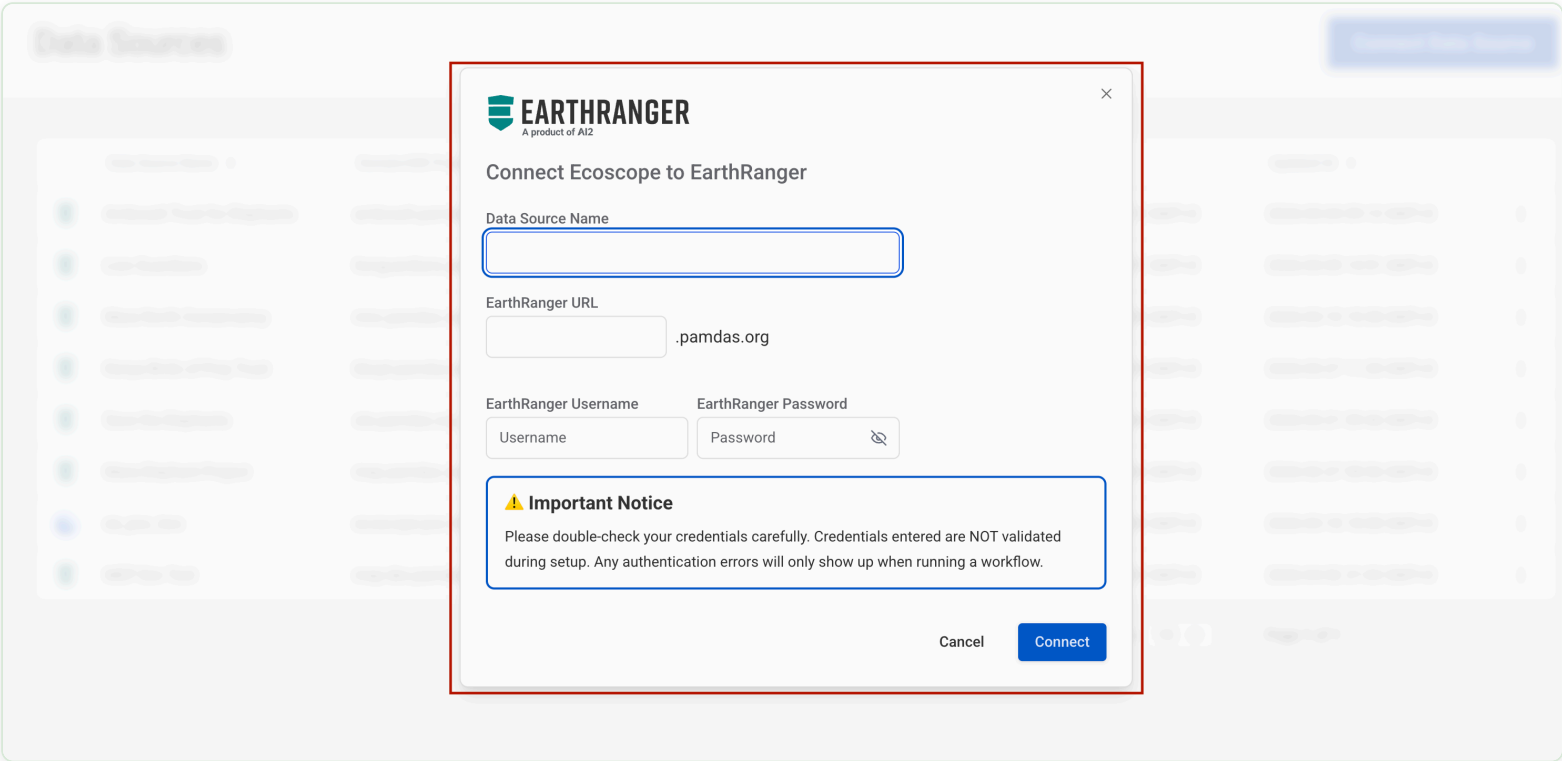
Navigate to **Data Sources** and click **Connect**, then select **EarthRanger**. Fill in the connection form:

Field	Description
Data Source Name	A label to identify this connection (e.g. `Mara North Conservancy`)

Field	Description
EarthRanger URL	Your instance URL (e.g. `your-site.pamdas.org`)
EarthRanger Username	Your EarthRanger username
EarthRanger Password	Your EarthRanger password

> Credentials are not validated at setup time. Any authentication errors will appear when the workflow runs.

Click **Connect** to save.

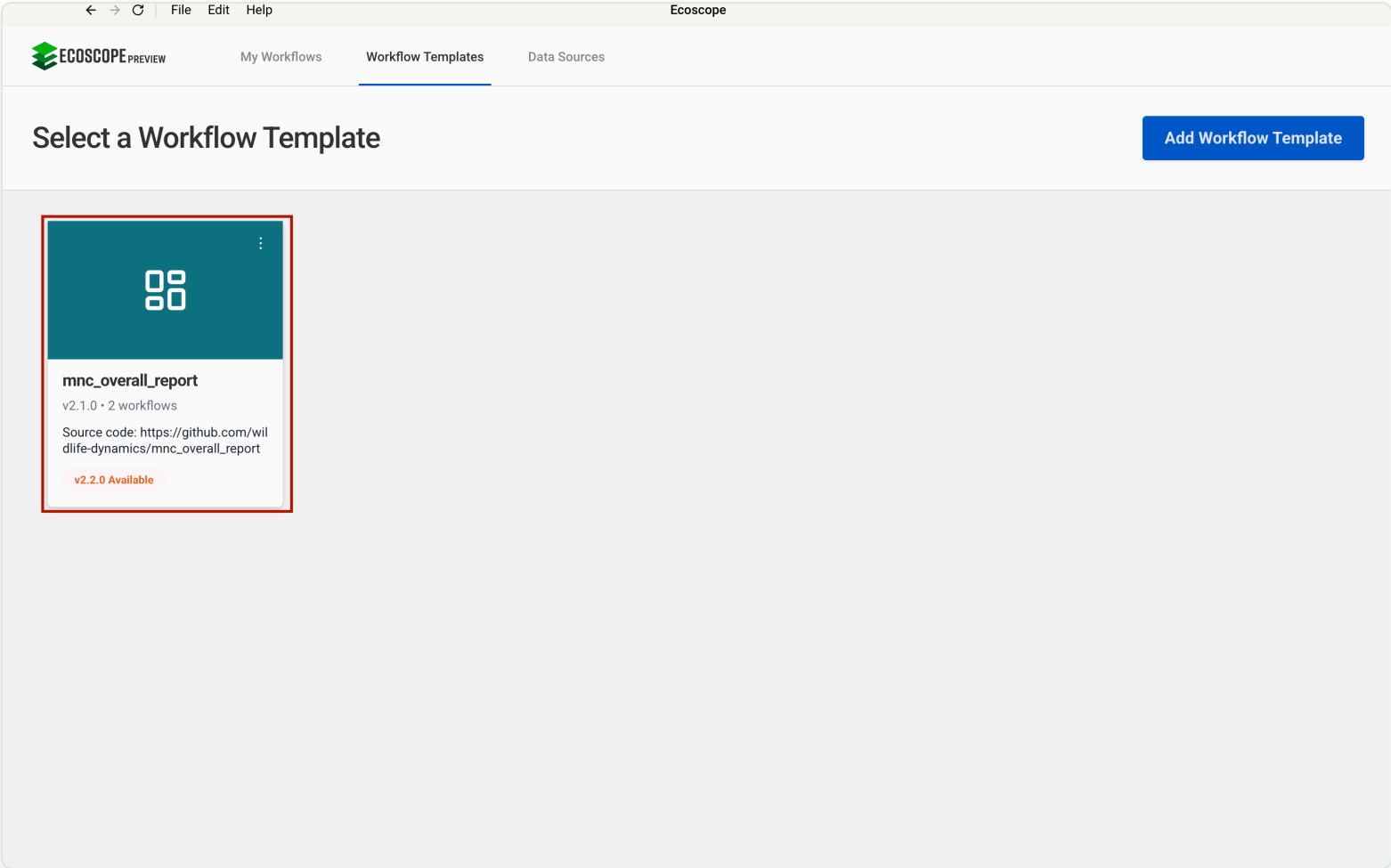


EarthRanger Connection

Step 3 — Select the Workflow

After the template is added, it appears in the **Workflow Templates** list as **mnc_overall_report**. Click the card to open the workflow configuration form.

> If a newer version is available (shown as **v2.2.0 Available** on the card), you can update the template before proceeding to pick up the latest changes.



Select Workflow Template

Step 4 — Configure Workflow Details, Time Range, and EarthRanger Connection

The configuration form has three sections on a single page.

Set workflow details

Field	Description
Workflow Name	A short name to identify this run
Workflow Description	Optional notes (e.g. reporting month or site)

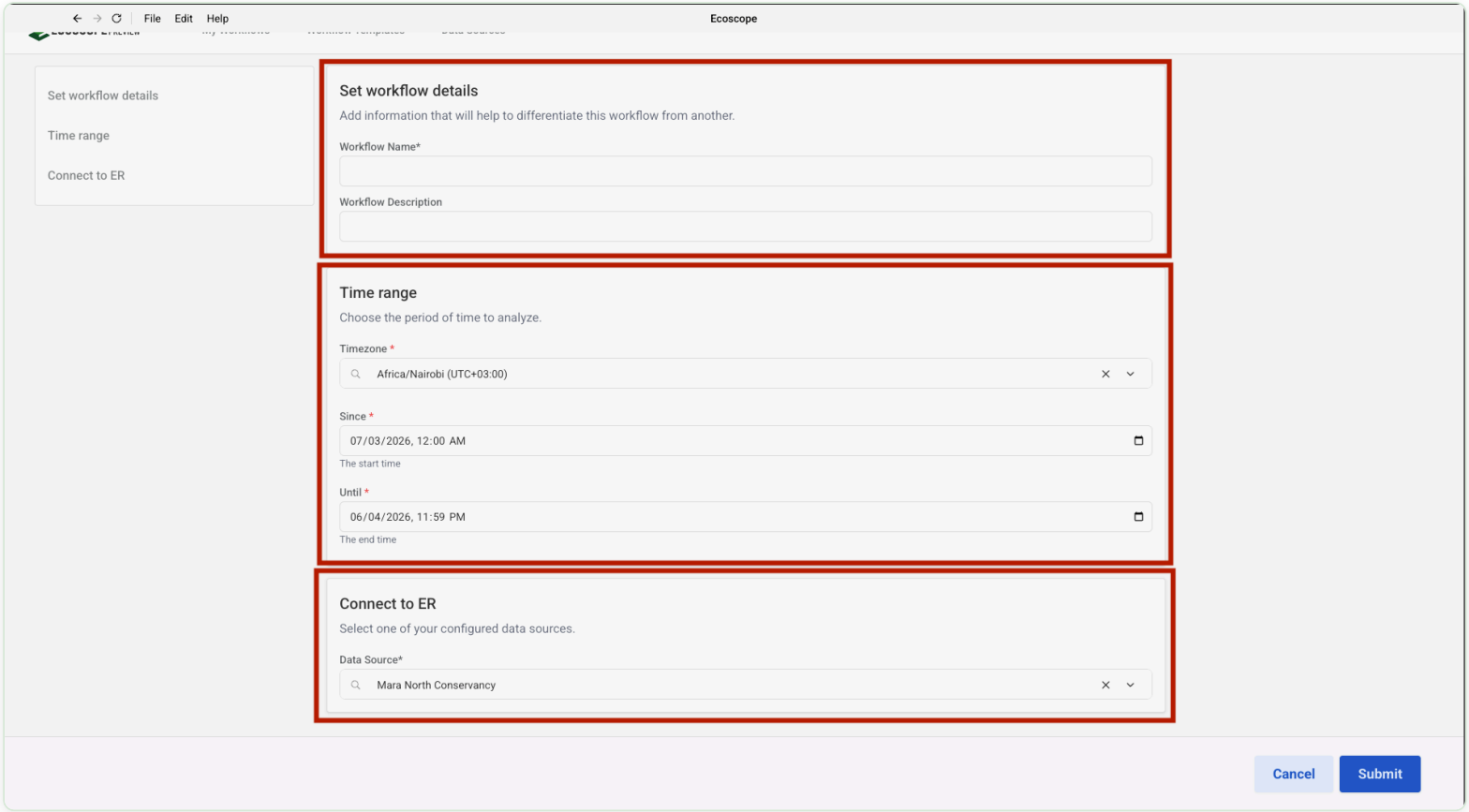
Time range

Field	Description
Timezone	Select the local timezone (e.g. `Africa/Nairobi UTC+03:00`)
Since	Start date and time — all data from this point is fetched
Until	End date and time of the analysis window

Connect to ER

Select the EarthRanger data source configured in Step 2 from the **Data Source** dropdown (e.g. `Mara North Conservancy`).

Once all three sections are filled, click **Submit**.



Configure Workflow Details, Time Range, and Connect to ER

Running the Workflow

Once submitted, the runner will:

1. Fetch weather station observations from the **ER2ER - From GMMF** subject group; extract seven meteorological variables; compute daily aggregates; generate weather charts.
2. Fetch all logistics events (balloon landings, airstrip operations, airstrip maintenance, airline complaints); clean and summarise each into a CSV table.

3. Fetch boma movement, cattle count, livestock predation, and illegal grazing events; generate maps and summary tables.
4. Fetch wildlife sighting events for elephant, buffalo, rhino, lion, leopard, cheetah, giraffe, and hartebeest; generate sighting maps, herd-size bar charts, and individual summary tables.
5. Fetch wildlife incident events; generate incident maps and summary tables.
6. Fetch foot, vehicle, and motorbike patrol observations; convert to trajectories; generate patrol maps, effort summaries, and overall coverage map.
7. Assemble all outputs into **overall_report.docx**.
8. Save all outputs to the directory specified by `ECOSCOPE_WORKFLOWS_RESULTS`.

Output Files

All outputs are written to `$ECOSCOPE_WORKFLOWS_RESULTS/`.

Weather

File	Description
`weather_summary_table.csv`	Daily per-station summary of all 7 weather variables
`precipitation_readings_over_time.html` / `.png`	Daily total precipitation
`temperature_readings_over_time.html` / `.png`	Daily mean surface air temperature
`wind_speed_readings_over_time.html` / `.png`	Daily mean wind speed
`wind_gusts_readings_over_time.html` / `.png`	Daily maximum wind gusts

File	Description
`soil_temperature_readings_over_time.html` / `.png`	Daily mean soil temperature
`relative_humidity_readings_over_time.html` / `.png`	Daily mean relative humidity
`atmospheric_pressure_readings_over_time.html` / `.png`	Daily mean atmospheric pressure

Logistics

File	Description
`balloon_landing_summary_table.csv`	Passenger records by balloon company and lodge
`airstrip_operations_summary_table.csv`	Client counts pivoted by camp/lodge and arrival/departure
`airstrip_maintenance_summary_table.csv`	Dated log of airstrip maintenance activities

Livestock and Boma

File	Description
`mobile_boma_movement_summary_table.csv`	Boma movement summary
`boma_movement_map.html` / `.png`	Boma movement locations map
`total_cattle_count_summary_table.csv`	Cattle count summary
`total_livestock_predation_summary_table.csv`	Livestock predation summary

File	Description
`livestock_predation_events.html` / `.png`	Livestock predation events map
`livestock_predation_summary_table.csv`	Livestock predation detail table
`illegal_grazing_map.html` / `.png`	Illegal grazing locations map

Wildlife

File	Description
`total_elephants_events_recorded.csv`	Elephant sighting event count
`elephant_sightings_events.html` / `.png`	Elephant sightings map
`elephant_herd_size_bar_chart.html` / `.png`	Elephant herd size distribution
`elephant_herd_types_map.html` / `.png`	Elephant herd types map
`total_buffalo_events_recorded.csv`	Buffalo sighting event count
`buffalo_sightings_events.html` / `.png`	Buffalo sightings map
`buffalo_herd_size_bar_chart.html` / `.png`	Buffalo herd size distribution
`buffalo_herd_types_map.html` / `.png`	Buffalo herd types map
`total_rhino_events_recorded.csv`	Rhino sighting event count
`rhino_sightings_events.html` / `.png`	Rhino sightings map
`total_lion_events_recorded.csv`	Lion sighting event count

File	Description
`individual_lions_summary.csv`	Individual lion records
`lion_sightings_events.html` / `.png`	Lion sightings map
`total_leopard_events_recorded.csv`	Leopard sighting event count
`individual_leopard_summary.csv`	Individual leopard records
`leopard_sightings_events.html` / `.png`	Leopard sightings map
`total_cheetah_events_recorded.csv`	Cheetah sighting event count
`individual_cheetah_summary.csv`	Individual cheetah records
`cheetah_sightings_events.html` / `.png`	Cheetah sightings map
`giraffe_sightings_events.html` / `.png`	Giraffe sightings map
`hartebeest_sightings_events.html` / `.png`	Hartebeest sightings map

Wildlife Incidents

File	Description
`wildlife_events_recorded.csv`	Wildlife incident event count
`wildlife_incidents_summary_table.csv`	Wildlife incidents summary
`wildlife_incidents_recorded_by_date.csv`	Wildlife incidents by date
`wildlife_incidents_map.html` / `.png`	Wildlife incidents map

Events Overview

File	Description
`total_events_recorded_by_date.csv`	All events recorded by date
`total_events_recorded_by_type.csv`	All events recorded by type
`total_events_recorded.html` / `.png`	All events chart

Patrols

File	Description
`patrol_purpose_summary.csv`	Patrol purpose breakdown
`foot_patrol_efforts.csv`	Foot patrol effort summary
`foot_patrol_trajectories.geoparquet`	Foot patrol trajectory data
`foot_patrols_map.html` / `.png`	Foot patrol trajectories map
`vehicle_patrol_efforts.csv`	Vehicle patrol effort summary
`vehicle_patrol_trajectories.geoparquet`	Vehicle patrol trajectory data
`vehicle_patrols_map.html` / `.png`	Vehicle patrol trajectories map
`motorbike_patrol_efforts.csv`	Motorbike patrol effort summary
`motor_patrol_trajectories.geoparquet`	Motorbike patrol trajectory data
`motorbike_patrols_map.html` / `.png`	Motorbike patrol trajectories map

File	Description
`overall_patrol_efforts.csv`	Combined patrol effort summary
`patrol_coverage.csv`	Patrol coverage data
`patrol_coverage_map.html` / `.png`	Overall patrol coverage map

Report

File	Description
`overall_report.docx`	Final Word report with all outputs assembled